

Practice Questions for Dual Problem

Generated by Scribe.AI

December 31, 2024

1 Questions

1. What is the fundamental difference between the worst-case and average-case performance of the simplex method, as discussed in the lecture slides?
 - a. The worst-case scenario always occurs in practice, while the average-case scenario is a theoretical ideal.
 - b. The worst-case performance is exponential, while the average-case performance is often observed to be polynomial.
 - c. The worst-case performance is determined by the size of the problem, while the average-case performance is independent of the problem size.
 - d. The worst-case and average-case performances are identical for all linear programming problems.
 - e. The worst-case performance is only relevant for very small problems, while the average-case performance is more relevant for large problems.
2. The Klee-Minty problem is a crucial example in the study of the simplex method. What key characteristic of this problem leads to its exponential worst-case time complexity?
 - a. The problem's objective function is highly non-linear.
 - b. The problem involves a large number of constraints compared to the number of variables.
 - c. The problem's feasible region is highly skewed and elongated, causing the simplex method to traverse many vertices.
 - d. The problem's constraints are highly correlated.
 - e. The problem's optimal solution is located at a vertex far from the origin.
3. Why is the number of iterations alone a poor criterion for evaluating the efficiency of a pivot rule in the simplex method?
 - a. The number of iterations is always the same regardless of the pivot rule used.
 - b. The number of iterations does not reflect the computational cost of each iteration.
 - c. The number of iterations is only relevant for large-scale problems.
 - d. The number of iterations is inversely proportional to the total computation time.
 - e. The number of iterations is not related to the problem size.

4. Smoothed analysis offers a different perspective on the simplex method's efficiency compared to worst-case analysis. What is the fundamental idea behind smoothed analysis?
 - a. Smoothed analysis focuses solely on the average-case performance of the simplex method.
 - b. Smoothed analysis considers the performance of the simplex method on problems with perfectly structured data.
 - c. Smoothed analysis examines the performance of the simplex method on problems where the input data has been slightly perturbed by random noise.
 - d. Smoothed analysis ignores the impact of pivot rules on the simplex method's performance.
 - e. Smoothed analysis only applies to linear programming problems with a small number of variables and constraints.

5. Consider the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $-x_1 + 3x_2 \leq 12$, $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$. What is the objective function of its dual problem?
 - a. Minimize $W = 12y_1 + 8y_2 + 10y_3$
 - b. Maximize $W = 12y_1 + 8y_2 + 10y_3$
 - c. Minimize $W = -y_1 + y_2 + 2y_3$
 - d. Maximize $W = -y_1 + y_2 + 2y_3$
 - e. Minimize $W = 3y_1 + 2y_2$

6. For the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$, what are the constraints of its dual problem?
 - a. $y_1 + 2y_2 \geq 3$, $y_1 - y_2 \geq 2$, $y_1, y_2 \geq 0$
 - b. $y_1 + 2y_2 \leq 3$, $y_1 - y_2 \leq 2$, $y_1, y_2 \geq 0$
 - c. $y_1 + 2y_2 \geq 3$, $y_1 - y_2 \geq 2$
 - d. $y_1 + 2y_2 \leq 3$, $y_1 - y_2 \leq 2$
 - e. $y_1 + 2y_2 = 3$, $y_1 - y_2 = 2$

7. Given the primal problem: Maximize $Z = c_1x_1 + c_2x_2$ subject to $a_{11}x_1 + a_{12}x_2 \leq b_1$, $a_{21}x_1 + a_{22}x_2 \leq b_2$, $x_1, x_2 \geq 0$. What is the correct expression for the dual problem's objective function?
 - a. Minimize $W = b_1y_1 + b_2y_2$
 - b. Maximize $W = b_1y_1 + b_2y_2$
 - c. Minimize $W = a_{11}y_1 + a_{21}y_2$
 - d. Maximize $W = a_{11}y_1 + a_{21}y_2$
 - e. Minimize $W = c_1y_1 + c_2y_2$

8. Formulate the dual problem for the primal problem: Maximize $Z = 2x_1 + 5x_2$ subject to $x_1 + 2x_2 \leq 10$, $x_1 + x_2 \leq 6$, $x_1, x_2 \geq 0$. What is the constraint related to the variable x_1 in the dual problem?
 - a. $y_1 + y_2 \leq 2$

- b. $y_1 + y_2 \geq 2$
 - c. $2y_1 + 5y_2 \leq 10$
 - d. $2y_1 + 5y_2 \geq 10$
 - e. $y_1 + y_2 = 2$
9. In the context of weak duality, if Z represents the objective function value of the primal problem and W represents the objective function value of the dual problem, which inequality always holds?
- a. $Z \geq W$
 - b. $Z \leq W$
 - c. $Z = W$
 - d. $Z > W$
 - e. $Z < W$
10. Consider the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $-x_1 + 3x_2 \leq 12$, $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$. What is the objective function of its dual problem?
- a. Minimize $W = 12y_1 + 8y_2 + 10y_3$
 - b. Maximize $W = -y_1 + y_2 + 2y_3$
 - c. Minimize $W = -12y_1 - 8y_2 - 10y_3$
 - d. Minimize $W = 12y_1 + 8y_2 + 10y_3$ subject to $-y_1 + y_2 + 2y_3 \geq 3$ and $3y_1 + y_2 - y_3 \geq 2$
 - e. Maximize $W = 12y_1 + 8y_2 + 10y_3$

2 Answers

1. What is the fundamental difference between the worst-case and average-case performance of the simplex method, as discussed in the lecture slides?
 - a. The worst-case scenario is a theoretical bound, not a guarantee of practical performance. The average-case performance is based on observations from many problem instances.
 - b. The Klee-Minty problem demonstrates that the simplex method can have exponential worst-case complexity. However, empirical studies and observations from practical applications suggest that the average-case performance is often polynomial, meaning the number of iterations grows polynomially with the problem size.
 - c. Both worst-case and average-case performances are influenced by problem size, but the relationship is different (exponential vs. often polynomial).
 - d. The worst-case and average-case performances are distinct and can differ significantly.
 - e. Both worst-case and average-case performances are relevant for problems of all sizes, but the average-case behavior is more indicative of typical performance.
2. The Klee-Minty problem is a crucial example in the study of the simplex method. What key characteristic of this problem leads to its exponential worst-case time complexity?
 - a. The objective function in the Klee-Minty problem is linear.
 - b. While the number of constraints can influence the runtime, the key factor in the Klee-Minty problem is the shape of the feasible region.
 - c. The Klee-Minty problem is specifically constructed to create a highly skewed and elongated feasible region. This forces the simplex method, using certain pivot rules, to visit an exponential number of vertices before reaching the optimal solution.
 - d. The constraints in the Klee-Minty problem are designed to create a specific geometric structure, not necessarily high correlation.
 - e. The location of the optimal solution is a consequence of the skewed feasible region, not the primary cause of the exponential runtime.
3. Why is the number of iterations alone a poor criterion for evaluating the efficiency of a pivot rule in the simplex method?
 - a. The number of iterations varies significantly depending on the pivot rule and the problem instance.
 - b. Different pivot rules may require different amounts of computation per iteration. A rule that reduces the number of iterations might still be less efficient if each iteration is computationally expensive.
 - c. The number of iterations is relevant for problems of all sizes, but the computational cost per iteration becomes more significant for larger problems.
 - d. There is no inverse relationship between the number of iterations and total computation time. A smaller number of iterations can still result in a longer total runtime if the iterations are more complex.
 - e. The number of iterations is directly related to the problem size and the chosen pivot rule.

4. Smoothed analysis offers a different perspective on the simplex method's efficiency compared to worst-case analysis. What is the fundamental idea behind smoothed analysis?
 - a. Smoothed analysis considers both worst-case and average-case scenarios, but it modifies the input data to make the analysis more realistic.
 - b. Smoothed analysis is designed to handle imperfections and randomness in real-world data, not perfectly structured data.
 - c. Smoothed analysis acknowledges that real-world problems are unlikely to be perfectly structured as in worst-case examples. It analyzes the performance of algorithms on problems where the input data has been slightly perturbed by random noise, providing a more realistic assessment of efficiency.
 - d. Pivot rules significantly impact the performance of the simplex method, and smoothed analysis considers this factor.
 - e. Smoothed analysis is applicable to linear programming problems of all sizes.

5. Consider the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $-x_1 + 3x_2 \leq 12$, $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$. What is the objective function of its dual problem?
 - a. The dual problem is formed by minimizing the objective function representing the constraints of the primal problem. The coefficients of the objective function in the dual problem are the right-hand side values of the constraints in the primal problem.
 - b. This is incorrect. The dual of a maximization problem is a minimization problem.
 - c. This is incorrect. These coefficients come from the primal problem's constraints, but they are not used in the dual objective function.
 - d. This is incorrect. The dual of a maximization problem is a minimization problem.
 - e. This is incorrect. These coefficients are from the primal objective function, but they are not directly used in the dual objective function.

6. For the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$, what are the constraints of its dual problem?
 - a. The constraints in the dual problem are derived from the coefficients of the variables in the primal problem's constraints. The inequalities are reversed because the primal problem is a maximization problem.
 - b. The inequalities should be greater than or equal to, not less than or equal to, because the primal problem is a maximization problem.
 - c. The non-negativity constraints $y_1, y_2 \geq 0$ are missing.
 - d. The inequalities should be greater than or equal to, not less than or equal to, because the primal problem is a maximization problem.
 - e. The constraints in the dual problem are inequalities, not equalities.

7. Given the primal problem: Maximize $Z = c_1x_1 + c_2x_2$ subject to $a_{11}x_1 + a_{12}x_2 \leq b_1$, $a_{21}x_1 + a_{22}x_2 \leq b_2$, $x_1, x_2 \geq 0$. What is the correct expression for the dual problem's objective function?
 - a. The dual objective function uses the right-hand side values (b_1, b_2) of the primal constraints as coefficients.

- b. The dual of a maximization problem is a minimization problem.
 - c. These coefficients are from the primal constraints but are not used in the dual objective function.
 - d. The dual of a maximization problem is a minimization problem.
 - e. These coefficients are from the primal objective function, but they are not directly used in the dual objective function.
8. Formulate the dual problem for the primal problem: Maximize $Z = 2x_1 + 5x_2$ subject to $x_1 + 2x_2 \leq 10$, $x_1 + x_2 \leq 6$, $x_1, x_2 \geq 0$. What is the constraint related to the variable x_1 in the dual problem?
- a. The inequality should be $\geq$, not $\leq$, because the primal problem is a maximization problem.
 - b. The dual constraint related to x_1 is derived from the coefficients of x_1 in the primal constraints. Since the primal is a maximization problem, the inequality in the dual is reversed.
 - c. This constraint is incorrect; it uses coefficients from the objective function.
 - d. This constraint is incorrect; it uses coefficients from the objective function and the inequality is reversed.
 - e. The dual constraints are inequalities, not equalities.
9. In the context of weak duality, if Z represents the objective function value of the primal problem and W represents the objective function value of the dual problem, which inequality always holds?
- a. This inequality would hold if we were discussing strong duality, which only applies at optimality. Weak duality holds for any feasible solutions to the primal and dual problems.
 - b. Weak duality states that the objective function value of the primal problem is always less than or equal to the objective function value of the dual problem, regardless of feasibility. This means that any feasible solution to the primal problem provides a lower bound for the optimal solution of the dual problem, and vice versa.
 - c. This equality only holds at optimality, when strong duality applies. Weak duality is a broader concept that applies to any feasible solutions.
 - d. This inequality is not always true. Weak duality states that $Z \leq W$ for any feasible solutions.
 - e. This inequality is not always true. Weak duality states that $Z \leq W$ for any feasible solutions.
10. Consider the primal problem: Maximize $Z = 3x_1 + 2x_2$ subject to $-x_1 + 3x_2 \leq 12$, $x_1 + x_2 \leq 8$, $2x_1 - x_2 \leq 10$, $x_1, x_2 \geq 0$. What is the objective function of its dual problem?
- a. The objective function of the dual problem is formed by taking the right-hand side values of the primal constraints as coefficients. Therefore, the objective function is Minimize $W = 12y_1 + 8y_2 + 10y_3$.
 - b. This incorrectly uses the coefficients from the primal constraints.
 - c. This uses the negative of the correct coefficients.
 - d. This includes constraints, which are part of the dual problem but not the objective function.
 - e. This is the correct form but should be a minimization problem.